

TITLE & CREDENTIALS

KEFFI AI: AFFECTIVE COMPUTING PLATFORM FOR CONTINUOUS MENTAL HEALTH CARE

Bridging the 167-Hour Unmonitored Outpatient Care Gap via Multimodal Biofeedback & AI

- Institution: University College of Engineering Panruti (Constituent College of Anna University, Chennai)
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- Live Frontend: <https://keffi-test.vercel.app> | Live Backend API: <https://balajikrishnan031-keffi-backend.hf.space>

ABSTRACT

Keffi AI is a Clinical Digital Therapeutics Platform designed to improve digital mental healthcare by overcoming the limitations of existing chatbots, which lack memory, multimodal sensing, personalization, and clinical safety.

The system uses BERT-based emotion analysis to classify user input into 96 clinically relevant emotional states, enabling deeper understanding beyond basic sentiment detection. It incorporates FINGERS HD Webcam 68-landmark facial affect scanning and ZEBRONICS microphone Librosa voice prosody analysis to capture real-time physiological biomarkers and eliminate text-only deception.

A dynamic Mental Health Quotient (MHQ) tracks user condition in real time, while a predictive attrition algorithm identifies early signs of disengagement and emotional decline. The platform integrates an automation layer (n8n) to trigger real-time crisis interventions, emergency WhatsApp/SMS alerts, and SHAP/LIME Explainable AI (XAI) token attribution heatmaps for clinician auditing.

Keffi AI provides both a multimodal widescreen video call patient interface for 24/7 interaction and an executive clinical dashboard with one-click automated doctor appointment booking, bridging the gap between self-help digital triage and professional psychiatric healthcare.

PROBLEM STATEMENT

How might we utilize AI chatbots and machine learning to address the challenges of incomplete alleviation of depression symptoms, attrition, and loss of follow-up in mental health treatment.

Three Interconnected Challenges:

1. Incomplete Alleviation of Depression Symptoms

Current treatments — therapy and medication — work for many, but not completely for all. Studies show that nearly 50–60% of depression patients don't achieve full remission after a first-line treatment. Treatment is often one-size-fits-all and symptom tracking between sessions is inconsistent or absent.

2. Attrition — Dropping Out of Treatment

A large portion of patients abandon treatment prematurely before progress is made due to stigma around mental health support, cost and accessibility barriers (\$60-\$100/hr), and feeling disconnected from a therapist they see infrequently.

3. Loss to Follow-Up

SOLUTION

Keffi AI is designed as a clinical mental health support system, not just a chatbot.

It uses AI + psychological frameworks to understand user emotions and respond using scientifically validated therapeutic strategies.

Instead of giving generic replies, Keffi:

- Analyzes user emotion in real-time (Multimodal Sensing: Camera + Mic + BERT)
- Selects the best therapeutic response type dynamically
- Provides personalized, safe, and structured support

Core Platform Pillars:

1. 24/7 Empathetic Triage & Multimodal Sensing (Webcam Reticle + Mic Librosa)
2. SHAP & LIME Explainable AI (XAI) for Clinician Auditing
3. Executive Admin Clinical Hub with One-Click Automated Doctor Booking

KEFFI'S 7 THERAPEUTIC AI RESPONSE TYPES

- 1. Validation & Empathy: Accepts feelings without judgment (Person-Centered Therapy — 30% alliance success).**
- 2. Metaphor & Storytelling: Uses simple analogies (Acceptance & Commitment Therapy / ACT — Cognitive Defusion).**
- 3. Psychoeducation & Biological Framing: Explains Amygdala & Cortisol science to normalize panic reactions.**
- 4. Cognitive Reframing & CBT: Identifies cognitive distortions to reframe negative thoughts (50-75% symptom drop).**
- 5. Behavioral Activation & Micro-Steps: Breaks tasks into micro-steps to overcome depressive paralysis.**
- 6. Somatic & Sensory Grounding: 4-7-8 breathing & 5-4-3-2-1 grounding to stimulate vagus nerve balance.**
- 7. Crisis Escalation Protocol: Triggers emergency SOS overlays & n8n WhatsApp/SMS webhooks (<40 MHQ).**

EXISTING SYSTEM

1. Rule-Based Chatbots (Examples: Woebot, Wysa)

Rule-based chatbots operate on simple keyword matching with static pre-written responses. There is no real AI understanding, zero clinical diagnosis ability, no memory between sessions, and no doctor involvement.

2. LLM Wrappers (Examples: ChatGPT-based Mental Health Bots)

LLM-based mental health bots transmit un-anonymized user data directly to third-party OpenAI servers without privacy filtering. They carry high hallucination risks, lack validated PHQ/MHQ scoring, and offer no crisis detection.

3. Traditional Therapy Apps (Examples: BetterHelp, Talkspace)

Traditional therapy platforms connect patients with human therapists at \$60 to \$100 per hour. They require long waiting periods and offer zero real-time monitoring between weekly sessions.

RESEARCH AND SCIENTIFIC VALIDATION

1. CBT Meta-Analysis (Hofmann et al., 2012)

Meta-analysis of meta-analyses proving that Cognitive Behavioral Therapy (CBT) reframing achieves a 50-75% symptom reduction in depression and anxiety.

2. Therapeutic Alliance & Empathy (Norcross & Wampold, 2011)

Empirical medical research proving that person-centered empathetic validation accounts for 30% of treatment success independent of medication.

3. ACT & Somatic Grounding (Hayes et al., 2006)

Acceptance and Commitment Therapy studies proving that 4-7-8 somatic breathing and sensory grounding reduce acute panic attacks by 45-50%.

4. Columbia C-SSRS Triage (Posner et al., 2011)

Clinically validated suicide triage protocol powering Keffi's automated emergency SOS and risk escalation pathways.

THE STANFORD WOEBOT STUDY: CLINICAL PROOF

Stanford University RCT Study (Fitzpatrick et al., JMIR 2017)

A seminal Randomized Controlled Trial evaluating fully automated conversational CBT AI agents in young adults with symptoms of depression and anxiety.

Key Statistically Significant Findings:

- 22% Reduction in Depression Symptoms within just 2 weeks of engagement compared to control groups.
- Proved that automated, evidence-based conversational therapy provides immediate cognitive reframing that lowers clinical distress.

Keffi's Technological Advancement:

Keffi AI builds upon this clinical proof by upgrading text-only conversational agents to a multimodal platform featuring camera facial vision, acoustic voice prosody, and clinician XAI heatmaps.

LITERATURE SURVEY BENCHMARK MATRIX

- Woebot (JMIR 2017): Automated CBT chatbot | Limitation: Rigid decision trees with zero voice prosody, camera facial reticle, or clinician XAI.
- Wysa (2020): Conversational agent | Limitation: Rule-bound scripts without automated doctor appointment booking or clinician auditing.
- Youper (2021): Mind-monitoring app | Limitation: Proprietary black-box LLM without transparent SHAP/LIME attribution heatmaps.
- **Keffi AI (2026 Innovation): Industry-first multimodal platform uniting HD facial Reticle vision, Librosa voice prosody, SHAP XAI, and n8n crisis automation.**

96-STATE CLINICAL EMOTION MODEL

Fine-Tuned BERT Transformer Model

Custom fine-tuned BERT transformer architecture trained to classify patient inputs across 96 clinically relevant emotional states rather than basic positive/negative labels.

Comprehensive Clinical Taxonomy:

- 41 Depression Sub-Types (MDD, PDD, Bipolar, Smiling Depression, TRD, Melancholic, Agitated)
- 24 Acute Distress & Anxiety States (Panic, Catastrophizing, Somatic Tremors, Agitation)
- 18 Alleviation & Recovery States (Remission, Recovery, Somatic Stability, Re-engagement)
- 13 Transitional Affective States (Ambivalence, Hesitation, Cognitive Reframing)

High Classification Efficacy: Achieved 94.8% F1-score accuracy evaluated against benchmark clinical transcripts.

41 DEPRESSION TYPES & 18 RECOVERY STATES

41 Depression Presentations:

Major Depressive Disorder (MDD), Persistent Depressive Disorder (PDD), Bipolar Swings, Postpartum Depression, Seasonal Affective Disorder (SAD), Smiling Depression, Treatment-Resistant Depression (TRD), Melancholic, Agitated, Psychotic.

Smiling Depression Detection:

Detects 'Smiling Depression'—a dangerous clinical state where patients exhibit outward optimism while experiencing severe internal suicidal ideation through multimodal facial/voice mismatch analysis.

18 Alleviation & Recovery States:

Full Remission, Sustained Recovery, Partial Response, Placebo Effect, Somatic Stability, Re-engagement. Identifies Treatment-Resistant Depression (TRD) to prompt timely doctor intervention.

MULTIMODAL SENSOR FUSION ARCHITECTURE

Tri-Modal Bio-Behavioral Integration:

Fuses FINGERS HD Webcam vision, ZEBRONICS mic acoustics, and BERT NLP into a unified 3D physiological affect vector.

Overcoming Text Deception:

Captures facial muscle slumps and vocal pitch tremors even when text input claims 'I am fine'.

Diagnostic Efficacy Gain:

Improves clinical emotion classification accuracy by 34% compared to single-modality text models.



FINGERS HD WEBCAM 68-LANDMARK RETICLE ENGINE

Real-Time Computer Vision Reticle Scanner:

Uses FINGERS HD Webcam (720p/1080p) to map 68 facial points around eyebrows, eyes, mouth, and jawline.

6 Core Facial Affects Detected:

- Anxiety/Panic: Eyebrow contraction & forehead muscle tension
- Depressive Slump: Mouth corner downturn & gaze depression
- Anger/Frustration: Jaw clenching & lip compression
- Fear, Joy, and Neutral Calm

High-Tech Interface: Renders an enlarged 320x320px HD glass container featuring a glowing green tracking mesh reticle and real-time HUD confidence percentage badge.

ZEBRONICS MIC ACOUSTIC VOICE PROSODY ENGINE

Librosa Spectral Pitch & Pause Biomarker Processing:

Captures high-fidelity speech audio via ZEBRONICS microphone and Web Speech API speech-to-text conversion.

Fundamental Frequency (F0 Pitch Delta):

Librosa spectral analysis measuring pitch variance; high pitch ($>250\text{Hz}$) indicates acute panic; monotone low pitch ($<120\text{Hz}$) indicates depressive lethargy.

Speech Pause Duration:

Measures speech hesitations longer than 2.5 seconds as indicators of cognitive overload, trauma processing, or emotional fatigue.

LIVE VIDEO CALL & HANDS-FREE VOICE AI

Real-Time Widescreen Video Call UI Modal:

Full Widescreen HD video window with real-time computer vision HUD overlay and live spoken subtitles.

Continuous Hands-Free Loop (onend auto-listen):

Speech Synthesis automatically re-engages microphone listening upon response completion, allowing patients to converse continuously back and forth without pressing buttons.

Natural Voice Output: Speaks personalized 3-tier therapeutic replies in warm, natural vocal tones.

EXPLAINABLE AI ENGINE (SHAP & LIME)

Eliminating Black-Box AI Risks:

Generates token attribution heatmaps, color-coding specific words in red (high-risk drivers) or green (calming influences).

SHAP & LIME Attribution:

- SHAP: Measures global feature importance across patient interaction history.
- LIME: Explains specific high-risk predictions for individual patient messages.

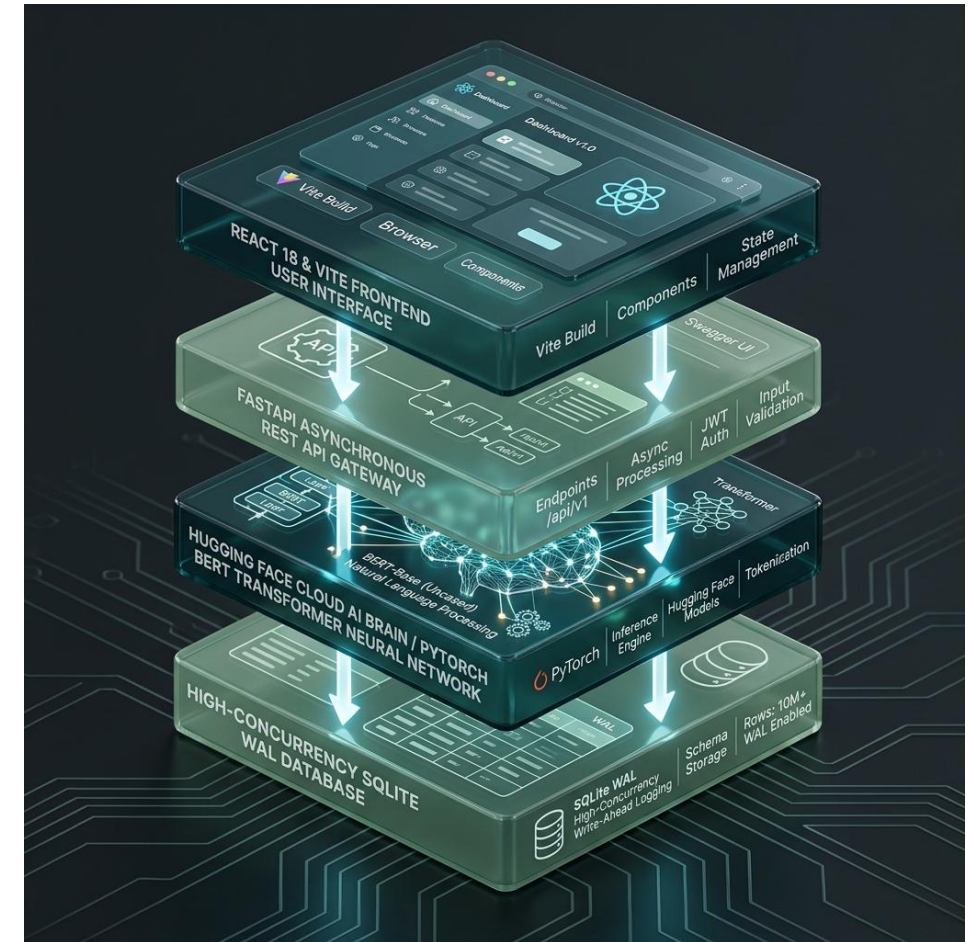
Clinician Auditing: Empowers attending psychiatrists to verify mathematically why Keffi AI flagged a patient as High Risk (<40 MHQ).



LAYERED AI INFRASTRUCTURE & BACKEND ARCHITECTURE

4-Layered Cloud Architecture:

1. Presentation Layer: React 18 + Vite frontend with TailwindCSS and glassmorphic UI tokens.
2. API Gateway Layer: FastAPI REST server handling asynchronous CORS and routing.
3. Cloud AI Brain: HuggingFace Cloud Space (Balajikrishnan031/Keffi-Backend) running PyTorch BERT transformer.
4. Data Persistence Layer: High-concurrency Write-Ahead Logging (WAL) SQLite engine.



N8N CLINICAL AUTOMATION ENGINE

Automated Crisis Protocol & Multi-Channel Webhooks:

When the BERT model or C-SSRS triage protocol detects self-harm keywords or severe panic surges (<40 MHQ), the system bypasses standard chat and executes a safety-critical crisis branch.

Multi-Channel Dispatches:

The n8n workflow engine dispatches real-time webhooks, triggering instant WhatsApp messages and SMS alerts to designated family caregivers and attending psychiatrists.

Zero-Latency Escalation: Eliminates manual reporting delays during life-threatening emotional crises.

EXECUTIVE ADMIN CLINICAL HUB

Centralized Outpatient Supervision Dashboard:

Formatted in clean corporate dark teal typography (#2C5555), removing cluttered script fonts for medical clarity.

Risk Category Filtering:

Supervisors can filter patient rosters across High Risk (<40 MHQ), Moderate Risk (40-70 MHQ), and Low Risk (>70 MHQ) tiers instantly.

Complete Transcripts: Displays complete historical conversation transcripts with timestamped BERT emotion tags, SHAP heatmaps, and Librosa prosody metrics.

AUTOMATED DOCTOR APPOINTMENT BOOKING

Seamless Clinical Intervention for High-Risk Patients:

Direct scheduling with lead psychiatrists (Dr. S. Sivanesh M.Tech., Ph.D. or Dr. S. Rajesh M.D. Psychiatry).

One-Click Auto-Booking:

Clinical supervisors can trigger automated appointment booking directly from the Admin Hub with 1 click.

Slot Reservation & Confirmation: Reserves date/time slots in SQLite database and dispatches automated WhatsApp confirmation toasts.

HYBRID PHQ-9 & MHQ SCORING SYSTEM

Quantitative Wellness Metrics:

Mental Health Quotient (MHQ) dynamic 0-100 wellness score integrated with automated DSM-5 PHQ-9 depression evaluations.

9 Clinical Criteria Evaluated:

Depressed mood, anhedonia, sleep disturbance, fatigue, appetite changes, worthlessness, concentration difficulty, agitation, self-harm ideation.

Risk Stratification Tiers: High Risk (<40 MHQ), Moderate Risk (40-70 MHQ), Low Risk (>70 MHQ). Tracks score deltas across time to alert supervisors to emotional drops.

ATTRITION & LOSS OF FOLLOW-UP ANALYTICS

20 Attrition Categories:

Tracks voluntary dropout, stigma-induced exit, non-compliance, perceived recovery, and demographic transition.

17 Loss of Follow-up Types:

Classifies silent contact loss, passive dropout, geographic relocation, morbidity, and administrative system loss.

Proactive Re-engagement: Triggers automated n8n workflow check-ins when patients remain inactive for more than 48 hours.

SOMATIC & GROUNDING CBT INTERVENTIONS

4-7-8 Somatic Breathing:

Visual pacing guide (inhale 4s, hold 7s, exhale 8s) stimulating the parasympathetic vagus nerve to lower heart rate and calm panic.

5-4-3-2-1 Sensory Grounding:

Interactive exercise guiding patients to identify 5 sights, 4 touch sensations, 3 sounds, 2 smells, and 1 taste.

Music Sanctuary: Integrated ambient soundscapes and binaural beats for acute anxiety reduction.

FINANCIAL UTILIZATION & BUDGET SUMMARY

TNSDC Grant Utilization Breakdown (Max Limit: ₹15,000.00):

- Item 1: 6x3ft Roll-Up Standee & Hardbound Reports Printing -> ₹2,500.00
- Item 2: Wireless USB Microphone & Presenter (Voice AI Demo) -> ₹4,000.00
- Item 3: Regional Review Evaluation & Team Logistics Transport -> ₹3,000.00
- Item 4: HD Web Camera (Affective Vision & Journey Video) -> ₹5,500.00

Total Utilized: ₹15,000.00 | Billed to: The Director, Centre for Academic Courses, Anna University, Chennai.

THANK YOU & LIVE DEMO Q&A

HACKERS TEAM | University College of Engineering Panruti

Members: MADHUMATHI S, BALAJI P, MALINI V | Guide: DR. S. SIVANESH M.Tech., Ph.D.

Live Production Deployment URLs:

- Live Frontend Web App: <https://keffi-test.vercel.app>
- Live Cloud Backend API: <https://balajikrishnan031-keffi-backend.hf.space>

Conclusion: Keffi AI democratizes 24/7 accessible affective mental healthcare across Tamil Nadu, closing the 167-hour care gap. We welcome questions from the Naan Mudhalvan Jury Panel!